Homework 2

Kaushik Vada

1) **1**. (10) Label the planes illustrated in Fig. 1, i.e. give the Miller indices.

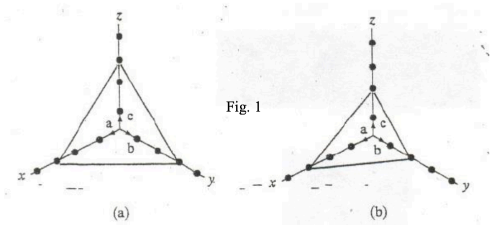


Fig. 1

a) $x = 3a$
$y = 3b$
$z = 3c$

$(3,3,3)$

invert: $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \times 3$

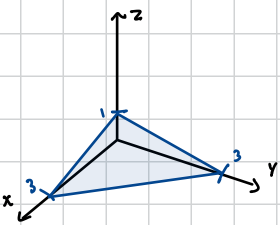
$(1,1,1)$

b) $x = 3a$, $y = 2b$, $z = 2c$

$(3,2,2)$ $\left(\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right) \times 6$

$(2,3,3)$

2) **2.** (10) Draw the (113) plane with its intercepts at the x, y, and z axes.



invert $\left(1, 1, \frac{1}{3}\right) \times 3$

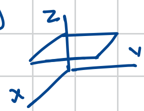
intercept $= (3,3,1)$

3) **3**. (10) Write down the Miller indices of the

(a) x-y plane.

(b) y-z plane.

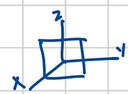
a)



Shifted by 1. Intercepts: $\infty, \infty, 1$

inverse: (001)

b)

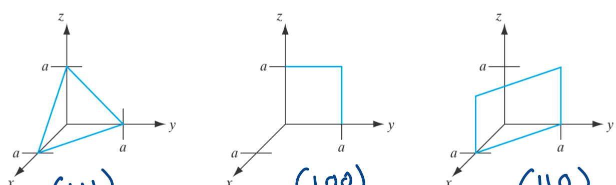


intercepts: $1, \infty, \infty$

invert: $1, 0, 0$

(100)

4) **4**. (10) Label the following planes using the correct notation for a cubic lattice of unit cell edge length *a*.

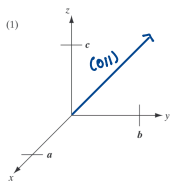


(111) (100) (110)

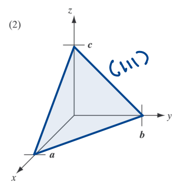
5)

5. (10) On the following sets of axes, (1) sketch the [011] direction and (2) a (111) plane (for a cubic system with primitive vectors **a**, **b**, and **c**).

(1)

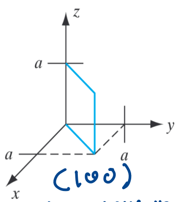


(2)

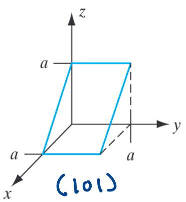


6)

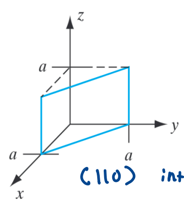
6. (a) (15) Determine the Miller indices for each plane below (shown within the first quadrant for 0 < x, y, z < a only, with the dotted lines for reference only).
(b) (10) Which of the planes are equivalent for a cubic crystal? Write down the Miller indices of the equivalent plane using the correct notation.



(100)
- lies entirely on the y-z plane



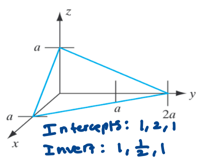
(101)
- hits x=a & z=a



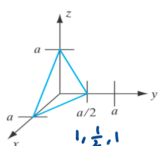
(110) intercepts (a, a, ∞)

7)

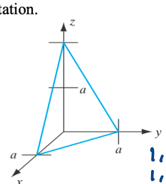
7. (15) Label each plane below with its Miller indices using correct notation.



Intercepts: 1, 2, 1
Invert: 1, $\frac{1}{2}$, 1
(212)



1, $\frac{1}{2}$, 1
(121)



1, 1, 2
1, 1, 1/2
(221)

8)

8. (5) What epitaxial growth method would you use to grow very thin (several nm), very precise layers of single-crystal semiconductor?

MBE (Molecular Beam Epitaxy) is used to grow very precise layers of single-crystal semiconductor

9)

9. (5) What growth method is used industrially to grow epitaxial, single-crystal layers of Si on Si?

The growth method that is used is **Chemical vapor deposition**

10) **10**. (5) Your boss tells you to get a p-type Si wafer out of the box. By looking at the wafer, how can you tell whether it is n-type or p-type?

A p-type wafer has one primary flat and also a second on a side that is adjacent to the primary flat.

A n-type wafer has always a primary and secondary flat on both ends of the wafer or the secondary flat 45 degrees adjacent to the primary flat.

11) **12**. (a) (5pts) Explain what is meant by an intrinsic semiconductor.

(b) (5pts) Draw the band diagram (E vs. x) at T=0K showing Ec, Ev and E_G. Show the electrons by filled circles and holes by empty circles.

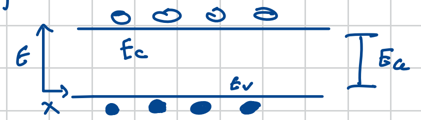
(c) (5pts) Repeat (b) for T=300K.

(d) (5pts) For intrinsic Si at room temperature, what are the electron and hole concentrations?

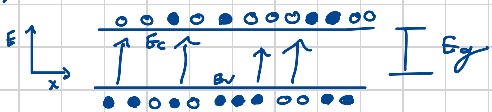
a) its a pure semiconductor w/ no intentional impurities

$n = p = n_i$

b) T = 0K



c) T = 300K



d) $n = p = n_i \approx \underline{1.0 \times 10^{10}\ cm^{-3}}$

13) **13**. (5pts) What is an extrinisic semiconductor?

$n \neq p$

14) **14**.(a) (5pts) Name 3 n-type dopants for Si. They are in which column of the periodic table?

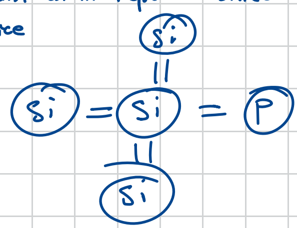
(b) (5pts) Name 1 p-type dopant for Si. It is in which column of the periodic table?

(c) (5pts) Explain with words and a sketch what is meant by substitutional doping.

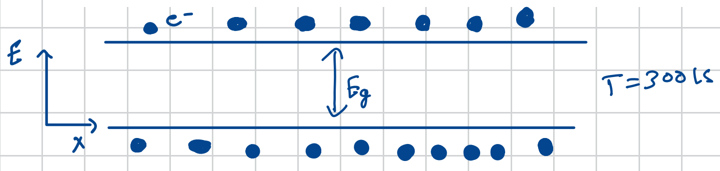
a) Phosphorus (P), Arsenic (As), Antimony (Sb)
(column 15)

b) Boron (B) (column 13)

c) Dopant atom replaces silicon atom in the crystal lattice



15) **15**. (5pts) On an energy band diagram (E vs. x), at room temperature, sketch the donor levels.
Indicate whether they are empty with empty circles or full with filled circles.
Similarly indicate any resulting holes in the valence band or electrons in the conduction band.
Indicate the charge of the donor with a + or – sign.



16) **16**. (5pts) Repeat (15) for acceptors.